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AIs
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ORF307: Optimization

Spring 2025 2^{nd} Midterm Exam Solutions

April 15, 11:00am – 12:20pm

Exam Rules

1. Please write out and sign the following pledge on the solution booklet: "I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this exam."
2. You are allowed a single sheet of paper, double-sided, hand-written or typed.
3. You cannot communicate with anyone during the exam.
4. All questions should be answered in the solution booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine. We will **not** grade anything written on the exam questions; we will only grade what you write in the solution booklet.
5. There are 3 problems worth a total of 100 points.
6. At the end of the exam, please turn in both your solution booklet and the exam questions.

1. *Problem 1 (35 points). Renewable Energy for AI Data Centers.*

A renewable energy farm is supplying power to a data center running large language models like ChatGPT. The farm needs to decide how much electricity to generate from its solar panels, wind turbines, and small nuclear reactor to maximize revenue while staying within resource constraints.

- Each megawatt (MW) of solar power generates \$40 in revenue and requires 3 hours of sunlight and 1 unit of maintenance.
- Each MW of wind power generates \$30 in revenue and requires 2 hours of wind and 2 units of maintenance.
- Each MW of non-renewable nuclear power generates \$50 in revenue and requires 0 hours of sunlight/wind but 3 units of maintenance and specialized handling.
- The farm has access to at most 180 hours of sunlight/wind and 100 units of maintenance per day.
- At least 30% of the energy generated should be from renewable sources.

Let x_1 be the amount of solar, x_2 be the amount of wind power, and x_3 be the amount of nuclear power generated, all in MW. Clearly, the farm cannot generate a negative amount of any power source.

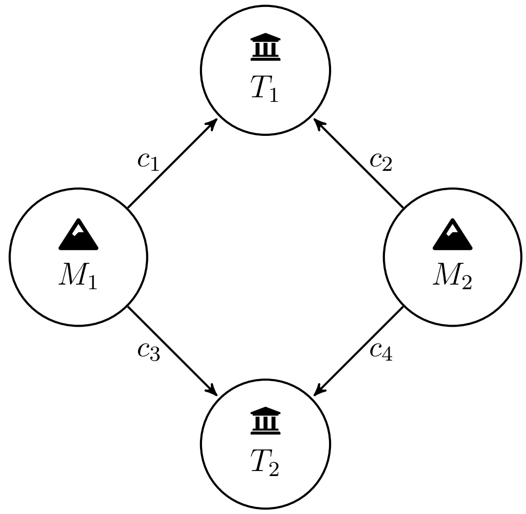
(a) (5 points) Write a linear program to maximize the farm's revenue subject to the above constraints.

(b) (10 points) Convert the LP from part (a) to standard form and construct a basic feasible solution with $(x_1, x_2, x_3) = (0, 0, 0)$. What is the basis B?

(c) (10 points) Carry out one iteration of the simplex method from the initial basic feasible solution and basis you constructed in part (b) using Bland's Rule. Report the values of the reduced costs vector $\bar{c}$, the basic direction d, the step size θ, the new basic feasible solution x^{new}, and the new basis B^{new}.

(d) (10 points) With the new basic feasible solution and basis from the end of part (c), compute the reduced costs vector $\bar{c}$. Has the method converged? What would you do next?

2. *Problem 2 (35 points). Moving Gold.*



You found gold in in two mines, M_1 and M_2, and your senior ORFE colleague Bob suggests you to move it (optimally!) to two treasuries T_1 and T_2. We denote the amount of gold in mine M_i as $a_i > 0$ for $i = 1, 2$, and the amount of gold that can be stored in treasury T_j as $b_j > 0$ for $j = 1, 2$. You can assume that $a_1 + a_2 = b_1 + b_2 = 20$. Due to safety concerns, transporting gold is very costly. The cost per pound to move gold from one of the mines to one of the treasuries is given by a cost vector $c = (c_1, c_2, c_3, c_4) = (2, 3, 4, 2) \in \mathbf{R}^4$ in the way that is specified in the diagram above. You would like to transport gold as cheaply as possible.

(a) (6 points) Formulate this problem as an LP in standard form.
Hint: The indices of the optimization variable $x \in \mathbf{R}^4$ should match the ones of c.

(b) (6 points) Formulate the dual problem.

Bob tells you that transporting gold with

$$\bar{x} = \frac{1}{20}(a_1 b_1, a_2 b_1, a_1 b_2, a_2 b_2),$$

is optimal. Let's see if he is right.

(c) (8 points) Verify that $\bar{x}$ is feasible for the primal LP.

(d) (7 points) If $\bar{x}$ was optimal, which constraints in the dual problem would be tight at the corresponding optimal dual variable $\bar{y}$?
Hint: use complementary slackness.

(e) (8 points) Is it possible to achieve the condition in (d)? Explain why and what this implies about $\bar{x}$ (and your friend Bob).
Hint: you should check if a specific linear system is solvable.

3. *Problem 3 (30 points). Final study rush in the reading period.*

You need to optimize the schedule of your final 5-day reading period. You are preparing for two subjects: Optimization and Machine Learning. Each subject has a different exam day and requires a different number of study days.

You can only study one subject per day. The exam happens at the end of the day, so you can still study the same subject on its exam day.

Subject	**Exam Day**	**Days Needed**
Optimization	3	2
Machine Learning	5	3

(a) (10 points) Formulate the problem as a max flow problem, with a source node s, sink node r, subject nodes v_1 (Optimization) and v_2 (Machine Learning), and nodes corresponding to individual days in your reading period $d = 1, \cdots, 5$. The flow corresponds to the amount of time studied (*i.e.*, a feasible schedule). The total flow from s to r is the total amount of time studied. Specifically write out *all* the nodes, arcs, costs c, and capacities u. You do not need to write the arc-node incidence matrix A.
Hint: Connect each subject node only to study days before the exam day.

(b) (5 points) Write the equivalent minimum cost flow problem.

(c) (5 points) Derive the dual problem of the minimum cost flow problem.

(d) (10 points) Say for the constraint $x_j \leq u_j$ where j are the arcs connecting s and subject nodes, we have associated dual variable y_j. What is the implication if the optimal $y_j^\star > 0$?